

Angel Chavira

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EDUCATION

University of Michigan- Ann Arbor, MI Aug 2020 – Jan 2026

Ph.D. in Nuclear Engineering (GPA: 3.97), M.S. in Nuclear Engineering (GPA: 3.96)

New Mexico Institute of Mining and Technology- Socorro, NM

Aug 2016 – May 2020

B.S. in Mechanical Engineering (GPA: 3.86)

TECHNICAL SKILLS

Codes & Software: LAMMPS, COMSOL, OVITO, SolidWorks, OpenMC, S/TRIM

Parallel Paradigms: OpenMP, OpenMPI, CUDA

Programming Languages: Python, C++, FORTRAN, MATLAB

Methods: molecular dynamics, (kinetic) Monte Carlo, finite element analysis, high-performance computing

Software Development: Git, GitHub, Conda, CMake, Make, Agile/Scrum

Debugging & Profiling Codes: TAU, DDT, GDB, MAP

RESEARCH EXPERIENCE

University of Michigan, Nuclear Engineering & Radiological Science Department

Ann Arbor, MI

Graduate Student Researcher, Computational Materials Group

Aug 2020 – Jan 2026

- Simulated defect evolution in electron irradiated α -iron at experimentally relevant dose rates using a novel atomistic method, benchmarking results against molecular dynamics and experiment. Key finding: C15 Laves phase clusters play a secondary role in immobile dislocation loop formation during irradiated material aging and hardening.
- Developed an atomistic off-lattice long-time dynamics code to simulate microstructural evolution in metals at extended timescales relevant to long-term nuclear materials reliability. Used saddle-point search swarms and defect optimization techniques to extend simulation timescales. Integrated LAMMPS for force evaluations and minimization functions.
- Designed and parallelized codebase for HPC deployment using OpenMPI and OpenMP. Successfully ran production simulations on Idaho National Lab's Sawtooth and Bitterroot, and University of Michigan's Great Lakes and Lighthouse computing clusters.
- Automated discovery of defect migration energy barriers for various defect cluster configurations using saddle-point search algorithms and the NEB method via LAMMPS and Python scripting. The energies required were used for the defect optimization technique.

University of Michigan, Nuclear Engineering & Radiological Science Department

Ann Arbor, MI

Graduate Student Researcher, Coupled Extremes Group

May 2023 – Jan 2026

- Characterized the multi-physics of irradiation-creep mechanisms in Fe and Ni systems under simultaneous thermal, stress, and radiation loading conditions. The aim was to provide deeper insights into creep-mode failure and radiation-induced segregation mechanisms in novel tapered specimen experiments.
- Carried out simulations of point defect kinetics, diffusion under stress/temperature gradients, and microstructural degradation using methods such as kinetic Monte Carlo and molecular dynamics.
- Performed 500+ point defect diffusion simulations to quantify anisotropic behavior and migration energies, finding diffusion was strongly influenced by defect type, clustering, crystal structure, and grain orientation under stress.
- Findings enhanced larger-scale creep-mode failure and radiation-induced segregation models to help solve unique microstructural observations in the tapered specimens.

Sandia National Laboratories

Albuquerque, NM

R&D Graduate Intern, Radiation Effects Theory

Jun 2022 – Sep 2022

- Contributed to verification and validation (V&V) of the Monte Carlo radiation transport code by profiling the performance of core cross-section evaluation routines.
- Optimized the radiation transport code to calculate interaction cross-sections for multiple elements simultaneously using OpenMPI.
- Created an SOP for code performance characterization to identify performance bottlenecks quickly.
- Participated in practical software engineering activities: worked in an Agile/Scrum development environment, followed team-implemented coding guidelines, and documentation practices.

New Mexico Tech, Mechanical Engineering Department

Socorro, NM

Undergraduate Researcher, Energetic Systems Research Group

Sep 2019 – Aug 2020

- Validated a novel tensile shock wave failure mode in aluminum under high-strain rate material conditions.
- Designed and ran molecular dynamics simulations of tensile shock wave fronts traveling through bulk aluminum.
- Produced data post-processing tools for visualizing density, velocity, and stress profiles as a function of time.

CONFERENCE PRESENTATIONS

The Minerals, Metals, and Materials Society (TMS), Columbus, OH, USA, 2025

Materials in Nuclear Energy Systems (MiNES), Las Vegas, NV, USA, 2025

Computer Simulation of Radiation Effects in Solids (COSIRES), Kingston, Canada, 2024

Materials in Nuclear Energy Systems (MiNES), New Orleans, LA, USA, 2023

Computer Simulation of Radiation Effects in Solids (COSIRES), Porquerolles, France, 2022

PUBLICATIONS

[1] A. Chavira, F. Gao, Microstructural Evolution in Irradiated Ferritic Iron: Dose Rate Effects on Defect Accumulation, *In Review, Science Advances*

[2] S. Lim, J. Kennedy, A. Chavira, M. Hirsch, T. Wei, D. Ryu, The Profile of Extreme Tension Wave Front in Aluminum, *Journal of Materials Science*, 2023

PROJECT EXPERIENCE

Jacobs Cryogenic Mechanical Testing

Aug 2019 – May 2020

- Developed a mechanical testing rig and procedure to evaluate the performance of carbon fiber coupons under cryogenic temperatures and various epoxy solutions for the Jacobs company.
- Designed and created a coupon containment system in compliance with strict OSHA cryogen safety standards.

NASA Telescope Truss Optimization

Aug 2018 – May 2019

- Conducted weight and resonant vibration optimization of NASA's space telescope truss design with COMSOL.
- Delivered a new truss design to NASA officials with a mass decrease of ~200 kg.